

AI 504 Problem set 1

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Due Friday, March 6

Instructions. Work in groups on the following problems. While it is okay to divide and conquer (and while I will not micromanage your group dynamics), *each of you is responsible for understanding every problem*. This means that you cannot simply blindly compile individual answers; you have to collaborate on a common submission.

Give sufficiently rigorous justification for all answers. Using generative AI, StackExchange, or any other online source of adaptive or problem-specific help is expressly prohibited. Scan your handwritten solution and/or develop solutions in LaTeX and submit as a pdf on itsLearning. Only one submission per group please!

Grading. A grade of 7 or higher means that you have *at understood all of the questions*. Even in the case that you do not completely solve a problem, write down your most coherent and intelligent response. (This could take the form of, e.g., a partial solution with known bugs, or an idea that you think will work but you cannot verify. Do not try to fake an answer, I will smell it.)

On top of which, a grade of 10 means a correct solution to at least half of the problems, and a grade of 12 to all of them. ("Correct solution" allows for minor typos and obviously fixable bugs, but all the ideas must be present.) Grades of 00, 2, or 4 will be awarded on an ad-hoc basis.

1. An *interval* is an ordered pair of real numbers (a, b) such that $a < b$. We say that the interval (a, b) *interleaves* the interval (c, d) in case $a < c < b < d$. Is the "interleaves" relation transitive? If not, what is its transitive closure?

2. Let X and Y be sets, let R and S be binary relations of type $X \times Y$, and let P and Q be subsets of X . Recall that the *forward image* $R(P)$ is the subset of Y obtained by following all R -arrows forwards out of P . (And similarly for all the other forward images in this problem.) For each of the following identities, either prove correct or provide a counterexample. *All proofs are short and all counterexamples are tiny.*

(a) $R(P \cup Q) = R(P) \cup R(Q)$

(b) $R(P \cap Q) = R(P) \cap R(Q)$

(c) $(R \cup S)(P) = R(P) \cup S(P)$

(d) $(R \cap S)(P) = R(P) \cap S(P)$

3. A *full binary tree* is a nonempty binary tree where each node is a leaf or has exactly two children. Define a function j from full binary trees to rational numbers by recursion as follows:

$$j(T) = \begin{cases} 1 & \text{if } T \text{ is a leaf} \\ 1 + \frac{j(T_0) + j(T_1)}{4} & \text{otherwise, where } T_0 \text{ and } T_1 \text{ are the two subtrees} \end{cases}.$$

Calculate the value of j on lots of small examples. Formulate a conjecture about how big it can get, and prove it.

4. Consider a directed graph G with 12 vertices that satisfies the following two properties:
- G is *strongly connected*, i.e., between any two distinct vertices u and v , there is a directed path from u to v as well as a directed path from v to u , and
 - there is at most one directed edge coming out of each vertex.

Draw G , and argue why your drawing is the *only possible* solution (up to isomorphism).